

Procedural Dungeon Roguelike

Turn-based roguelike with wave-function-collapse level generation, fully deterministic seeds, and a data-driven ability system.

Godot	C#	GDScript	WFC
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VERSION	STATUS	STAGE	ENGINE	PLATFORMS	UPDATED
0.7.1	In Progress	Vertical slice	Godot 4.4	PC, Steam Deck	Sep 2026

1. Overview

A turn-based roguelike whose dungeons are generated by wave-function collapse over a tile constraint set rather than by room-and-corridor placement. The solver produces layouts that obey authored adjacency rules — a forge never opens directly onto a flooded vault, corridors always resolve to something — which gives the generated space a hand-made grammar without hand-made floors.

Everything downstream of the seed is deterministic. Given a seed and a run of inputs, the entire run replays identically: the same layout, the same loot, the same enemy decisions. That makes daily challenge runs possible, makes bug reports reproducible from a seed and input log alone, and lets balance changes be evaluated by replaying a corpus of recorded runs.

2. Generation pipeline

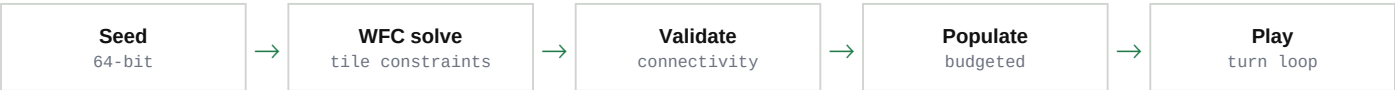


Figure 1 — A failed validation rewinds to the solver with the same seed and a recorded attempt counter, keeping generation deterministic across retries.

The solver works on a tile grid with a constraint table generated from authored sample chunks. When collapse contradicts, it backtracks to the last decision rather than restarting, which keeps generation under the frame budget for a 64×64 floor. Validation then checks that the exit is reachable, that every locked door has its key on the correct side, and that the critical path length falls within the floor's authored range.

3. Systems

RNG	Splittable deterministic PRNG; generation, loot, and AI each draw from a named stream.
WFC solver	Constraint propagation with backtracking over an authored tile adjacency table.
Validator	Connectivity, lock-and-key ordering, and critical path length checks with rewind.
Populator	Spends a per-floor encounter and loot budget against the room graph's tags.
Ability system	Data-driven effects composed from typed primitives; no per-ability script.
Turn scheduler	Energy-based ordering; actions cost energy, speed sets accrual rate.

4. Current status

AREA	STATE	NOTES
Generation and validation	Working	Three biomes with authored tile sets; stable at 64×64.
Ability system	Working	48 abilities defined; effect primitives considered complete.
Enemy AI	In progress	Utility scoring in place; pack coordination unimplemented.
Meta progression	Not started	Run history recorded but nothing unlocks from it yet.
Audio and VFX	Placeholder	Programmer art and stub sounds throughout.

5. Ability definitions

```
{
  "id": "frost_lance",
  "cost": { "energy": 100, "focus": 2 },
  "target": { "shape": "line", "range": 6, "pierces": true },
  "effects": [
    { "type": "damage", "amount": "2d6 + int", "element": "frost" },
    { "type": "status", "id": "chilled", "turns": 3,
      "unless": { "tag": "frost_immune" } },
    { "type": "terrain", "id": "ice", "radius": 0, "turns": 5 }
  ],
  "tags": ["spell", "frost", "ranged"],
  "ai_hint": { "prefers": "clustered", "min_targets": 2 }
}
```

Listing 1 — Abilities are data. Enemies and the player share the same executor, so anything an enemy can do is castable.

6. Determinism rules

- **No floating point in simulation.** Damage, positions, and timers are integers or fixed point. Floats exist only in presentation, where divergence cannot affect the run.
- **Stable iteration order.** Entities are processed by a stable identifier, never by hash or dictionary order, so the same turn resolves identically across platforms and builds.
- **Named streams.** Adding a draw to the AI stream cannot shift loot or generation, which keeps previously recorded seeds valid across patches as long as the constraint tables are unchanged.

7. Known issues

- Backtracking depth spikes on the flooded biome when the water tile set is combined with narrow corridors, occasionally pushing floor generation past 200 ms.
- Replay divergence appears after roughly 40 minutes of recorded play, traced to a status-effect expiry that reads presentation time rather than the turn counter.
- Steam Deck input remapping is incomplete; the ability radial assumes a keyboard modifier that has no gamepad equivalent.

8. Remaining work

MILESTONE	SCOPE	TARGET
Slice complete	Pack AI, replay divergence fixed, gamepad parity.	Q4 2026
Content build	Six biomes, boss encounters, meta progression loop.	Q1 2027
Early access	Daily seeds, run history, audio and VFX pass.	Q2 2027